

CS 2300 HW1

- 1) Translate the following English sentences into logic. First, define your basic propositions and use logical operations to connect them.
- (a) You pass the class only if you do homework or attend class regularly.
 - (b) I will play tennis if it is sunny and not windy.
 - (c) You will pass the class if and only if you attend class regularly and do your home- work

Ans) P: Passing the class
H: doing the homework
C: Attending class regularly
T: Play tennis
W: It is windy

- a) $P \rightarrow (H \vee C)$
- b) $T \rightarrow \text{Not windy}$
- c) $P \leftrightarrow (H \text{ and } C)$

- 2) State the converse, inverse and contrapositive of the statement “if it is Friday night, I will party.” Your statements should be in English.

Ans) **Converse:** If I party, then it is a friday night.

Inverse: If it is not a friday night, then I will not party.

Contrapositive: If I am not partying, then it is not a friday night

3) Determine whether $\leftrightarrow$ is associative, i.e., whether $(p \leftrightarrow q) \leftrightarrow r$ is equivalent to $p \leftrightarrow (q \leftrightarrow r)$. Prove your answer using a truth table.

p	q	r	$p \leftrightarrow (q \leftrightarrow r)$	$(p \leftrightarrow q) \leftrightarrow r$
T	T	T	T	T
T	T	F	F	F
T	F	T	F	F
T	F	F	T	T
F	F	F	F	F
F	F	T	T	T
F	T	F	T	T
F	T	T	F	F

4) $\text{NOT}(p) \rightarrow q = (p \text{ OR } q)$

$\text{NOT}(q) \rightarrow r = (q \text{ OR } r)$

Thus L.H.S = $(p \text{ OR } q) \text{ AND } (q \text{ OR } r)$

$(\text{NOT}(p) \text{ OR } \text{NOT}(r)) \rightarrow q = \text{NOT}(\text{NOT}(p) \text{ OR } \text{NOT}(r)) \text{ OR } q$ (implication rule)

$= (p \text{ AND } r) \text{ OR } q$

$(p \text{ OR } q) \text{ AND } (q \text{ OR } r) = (p \text{ AND } (q \text{ OR } r)) \text{ OR } (q \text{ AND } (q \text{ OR } r))$

$q \text{ AND } (q \text{ OR } r) = q$

$(p \text{ AND } (q \text{ OR } r)) \text{ OR } q$

$(p \text{ AND } q) \text{ OR } (p \text{ AND } r) \text{ OR } q$

$= q \text{ OR } (p \text{ AND } r)$ (Absorption Law)

$= \text{R.H.S}$ (Right Hand Side)

Steps	Reasoning
$(\text{NOT}(p) \rightarrow q) \text{ AND } (\text{NOT}(q) \rightarrow r)$	Given Expression
$(\text{NOT}(p) \text{ OR } q) \text{ AND } (\text{NOT } q \text{ OR } r)$	Implication rule ($a \rightarrow b = \text{NOT}(a) \text{ OR } b$)
$(\text{NOT}(p) \text{ AND } \text{NOT}(q)) \text{ OR } (\text{NOT}(p) \text{ AND } r) \text{ OR } (q \text{ AND } \text{NOT}(q)) \text{ OR } (\text{NOT}(q) \text{ AND } r)$	Distributive Property
$(\text{NOT}(p) \text{ AND } \text{NOT}(q)) \text{ OR } (\text{NOT}(p) \text{ AND } r) \text{ OR } (q \text{ AND } r)$	Since q and $\text{NOT}(q)$ is always false, so we remove that.
$\text{NOT}(p) \text{ AND } (\text{NOT}(q) \text{ OR } r) \text{ OR } (q \text{ AND } r)$	Factoring $\text{NOT}(p)$

NOT(p) AND NOT(q) OR NOT(p) AND r OR q AND r	Using Distributive property for rewriting the equation.
(NOT(p) OR NOT(r)) $\rightarrow$ q	Implication rule ($a \rightarrow b = \text{NOT}(a) \text{ OR } b$)
Conclusion: (NOT(p) $\rightarrow$ q)) AND (NOT(q) $\rightarrow$ r) and (NOT(p) OR NOT(r)) $\rightarrow$ q are equivalent	

- 5) A: Alice watches Wicked
 B: Bob watches Wicked
 C: Cindy watches Wicked
 D: Don watches Wicked
 E: Ellen watches Wicked

Meaning of each statement:

- 1) Either Alice or Bob, or both, watch Wicked. ($A \text{ or } B$)
- 2) Cindy and Don watch different movies. ($\text{Not } C \rightarrow D$)
- 3) If Ellen watches Wicked, then so does Cindy. ($E \rightarrow C$)
- 4) Alice and Don watch the same movie. ($A \leftrightarrow D$)
- 5) If Bob watches Wicked, then so do Alice and Ellen. ($B \rightarrow (A \text{ and } E)$)

Case 1: Bob watches Wicked

- 1) From ($B \rightarrow (A \text{ and } E)$) we can say that if Bob watches Wicked then so Alice and Ellen.
- 2) From the statement above and ($E \rightarrow C$) we can say that Cindy also watches Wicked
- 3) From the statement above and ($\text{Not } C \rightarrow D$) we can say that Don watches Brutalist.
- 4) From ($A \leftrightarrow D$) we can say that Don also watches Wicked since Alice watches Wicked as well, which is a false statement, so that means Bob has not watched Wicked.

Case 2: Bob watches Brutalists

- 1) From ($A \text{ or } B$) we can say that Alice watches Wicked, since Bob watches Brutalists
- 2) From ($A \leftrightarrow D$) we can say that Don watches Wicked as well
- 3) From ($\text{Not } C \rightarrow D$) we can say that Cindy watches Brutalists
- 4) From ($E \rightarrow C$) we can say that Ellen watches Brutalists

Thus, from Case 2 we can say that Alice and Don watch Wicked while Cindy, Ellen and Alice watch Brutalists.

6) $p \text{ XOR } q = (p \text{ AND } (\text{NOT}(q)) \text{ OR } ((\text{NOT } p) \text{ AND } q)$

If $q = \text{true}$, then we can say that:

$p \text{ XOR TRUE} = (p \text{ AND NOT(TRUE)}) \text{ OR } (\text{NOT}(p) \text{ AND TRUE}))$

$p \text{ XOR TRUE} = \text{NOT}(p)$

$p \text{ OR } q = \text{NOT}(\text{NOT}(p) \text{ AND } \text{NOT}(q))$ (De Morgan's law)

$p \text{ OR } q = (p \text{ XOR TRUE}) \text{ AND } (q \text{ XOR TRUE})$

Thus, we can say that $\{\text{XOR}, \text{AND}, \text{TRUE}\}$ is functionally complete.

7 a)

p	q	r	$p \text{ OR } q$	$\text{NOT}(q) \text{ OR } r$
T	T	T	T	T
T	T	F	T	F
T	F	T	T	T
F	F	F	F	T
F	T	T	F	T
F	T	F	T	F
T	F	F	T	T
F	F	T	F	T

b) From the truth table we get the following:

1. $(p \text{ and } q \text{ and } r)$
2. $(p \text{ and } \text{NOT}(q) \text{ and } r)$
3. $(\text{NOT}(p) \text{ and } \text{NOT}(q) \text{ and } \text{NOT}(r))$
4. $(\text{NOT}(p) \text{ and } q \text{ and } r)$
5. $(p \text{ and } \text{NOT}(q) \text{ and } \text{NOT}(r))$
6. $(\text{NOT}(p) \text{ and } \text{NOT}(q) \text{ and } r)$

Thus DNF: $(p \text{ AND } q \text{ AND } r) \text{ OR } (p \text{ AND } \text{NOT}(q) \text{ AND } r) \text{ OR } (p \text{ AND } \text{NOT}(q) \text{ AND } \text{NOT}(r)) \text{ OR } (\text{NOT}(p) \text{ AND } q \text{ AND } r) \text{ OR } (\text{NOT}(p) \text{ AND } \text{NOT}(q) \text{ AND } r) \text{ OR } (\text{NOT}(p) \text{ AND } \text{NOT}(q) \text{ AND } \text{NOT}(r))$

c) The proposition is false in these 2 cases (propositions 2 and 6) and can be represented as:

- 1) $(\text{NOT } p \text{ or } \text{NOT } q \text{ AND } r)$
- 2) $(p \text{ OR } \text{NOT } q \text{ AND } r)$

Thus CNF: $(\neg p \vee \neg q \vee r) \wedge (p \vee \neg q \vee r)$